

CSE343/CSE543/ECE363/ECE563: Machine Learning (UG/PG)
Winter 2025

Quiz-1 (12 points)

Roll number:

Duration: 60 minutes

Instructions

- **Institute's Academic Dishonesty Policy is Applicable.**
- All questions are compulsory.
- There can be multiple ways to approach a question. Justify all your answers appropriately. Questions without justification will get zero marks.
- Students will not be allowed to use the washroom during the test. Any request in this regard will not be entertained.
- Any student trying to copy during the test will be awarded a grade of F in the course along with his accomplice.

1. **[CO1, CO2] Class Imbalance** **(2 points)**

Consider highly imbalanced data for the problem of binary classification.

- (a) Discuss how class imbalance may influence the performance of a KNN classifier. (1 points)
- (b) What techniques can be used to address potential biases arising from this imbalance? (1 points)

2. **[CO2, CO4] Bias and Variance** **(2 point)**

Given the following options: High/Low Bias, Underfitting, Overfitting, Good Fit, fill the table with the correct matching options. A cell may have more than one answer

Train Set Error	13.5%	12%	3%	1%
Val Set Error	25%	13%	16%	1.5%
Bias	.	.	.	.
Underfitting / Overfitting	.	.	.	.

3. **[CO1, CO2] CNNs** **(2 point)**

Given an RGB image of size 224×300 , a Convolutional Neural Network (CNN) is applied with two convolutional layers: the first layer has 16 kernels of size 5×5 , stride 1, and padding 0, while the second layer has 32 kernels of size 3×3 , stride 2, and padding 0. Compute the output feature map size after each convolutional layer and determine the total number of trainable parameters in the CNN.

4. **[CO1, CO2] Gradient Descent** **(2 point)**

With learning rate $\eta = 0.003$ and $W_{old} = 0.897$ derive and write new updated weight W_{new} using this loss function equation -

$$L(x, \theta) = \left(\frac{1}{Y_{actual} - Y_{predicted}} \right)^2.$$

Hint: Loss L is not differentiable when the denominator is zero. Since the actual loss would be zero when predicted value of y is equal to actual y , you do not require to take the derivative at that point.

5. [CO1, CO4] Ridge Regression

(2 point)

Mention which statements are True about Ridge Regression and justify your answer also. Without justification, there will be no marks.

- (a) A way to perform feature selection, as ridge regression encourages weights to be exactly zero.
- (b) In ridge regression, increasing the regularization parameter λ , increases the reliance of the model on the data and vice versa.

6. [CO1, CO2] Convolutional Neural Networks

(2 point)

Consider the below CNN architecture - In the first layer, we have a one-dimensional convolution with a

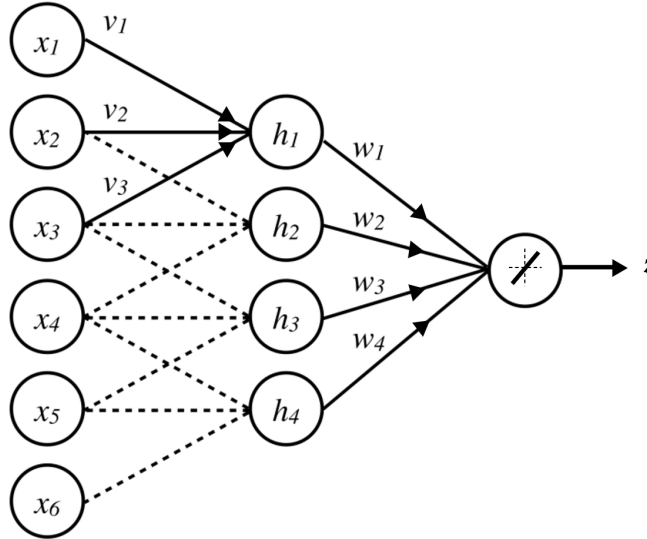


Figure 1: CNN Architecture

single filter of size 3 such that $h_i = s\left(\sum_{j=1}^3 v_j x_{i+j-1}\right)$. The second layer is fully connected, such that $z = \sum_{i=1}^4 w_i h_i$. The hidden units' activation function $s(x)$ is the logistic (sigmoid) function. The output unit is linear (no activation function). We perform gradient descent on the loss function $R = (y - z)^2$, where y is the training label for x .